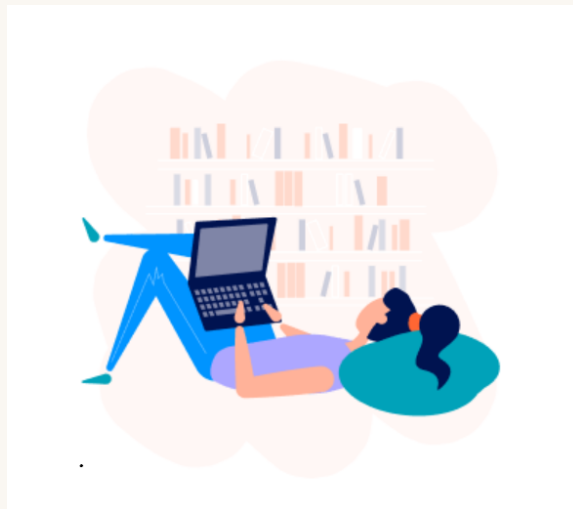


CareerPilot AI

From career confusion to a clear, evidence-based path toward employment.



**A career readiness workspace for students, graduates,
and early-career professionals.**

1. Executive Summary

CareerPilot AI helps students understand why they are unemployed and what they should do to become job-ready. Instead of offering generic courses or motivational advice, the platform compares a student's current profile with the expectations of a selected role and company. It turns identified gaps into a practical roadmap with daily learning, portfolio work, assessments, interview preparation, and recruiter-style feedback.

The project focuses on the gap between education and employability. Students may have degrees, certificates, and theoretical knowledge, but employers look for evidence: real projects, problem-solving, deployment, communication, and the ability to explain technical decisions. CareerPilot connects market reality, personal diagnosis, execution, and proof in one accessible workspace.

Core promise

Where am I now? Why am I being rejected? What does my target company expect? What should I learn, build, and practice next?

2. Problem and Audience

Students in Pakistan and other emerging markets often learn without a clear target. They do not know which skills matter for a specific job, why applications fail, what projects employers value, or how to prepare for interviews. Students outside major cities may also lack access to mentors, professional networks, and expensive career programs.

CareerPilot serves university students, recent graduates, self-taught developers, career switchers, and early-career professionals. It is also relevant to universities and training organizations that want a structured employability journey for their learners.

3. Solution and User Journey

CareerPilot converts uncertainty into a sequence of decisions and actions:

01 | Understand

Onboarding captures education, experience, skills, projects, certifications, and career goals.

02 | Diagnose

The platform identifies missing skills, weak proof, outdated knowledge, and rejection risks.

03 | Compare

Market Reality connects the student with role, company, Pakistan, and global hiring expectations.

04 | Execute

A personalized roadmap provides daily learning, practical tasks, and portfolio-focused projects.

05 | Practice

MCQs, interview preparation, and recruiter simulation turn knowledge into confident evidence.

06 | Progress

Assessments, notifications, readiness, and completion tracking make progress visible.

4. Key Capabilities

- Personalized career diagnosis with role and company context.
- Pakistan-focused Market Reality covering employer expectations and practical opportunities.
- Day-by-day roadmap with learning explanations, hands-on tasks, and project evidence.

- Scenario-based MCQ assessments with scores, explanations, and attempt history.
- Interview preparation for technical, behavioral, project-defense, and company-specific questions.
- Recruiter Audit that explains how a student's profile may be screened.
- Progress tracking, notifications, roadmap completion, and certificate unlocking.
- Responsive experience for desktop and mobile users.

5. Innovation

CareerPilot is not only a course platform, chatbot, resume tool, or job board. Its innovation is the connection between diagnosis, market reality, learning execution, assessment, interview preparation, recruiter feedback, and proof. The platform helps a student move from knowing about a skill to demonstrating that skill in a way an employer can understand.

6. Technology and Architecture

The working prototype uses a modern full-stack architecture designed for maintainability, security, and future scale. The client uses React, TypeScript, TanStack Router, TanStack Query, Tailwind CSS, and Motion. Supabase provides authentication, database persistence, user profiles, career goals, skills, projects, resumes, roadmaps, assessments, notifications, and recruiter sessions.

Server functions protect sensitive operations and keep AI credentials away from the browser. The AI layer supports Groq, Gemini, and OpenRouter providers with fallback handling. Market and technology reports are cached to reduce repeated requests and improve loading performance. Server-side validation protects assessment results and user-scoped data.

Layer	Implemented approach
Experience	Responsive React screens, mobile navigation, dashboard, roadmap, diagnosis, interview, recruiter, resume, and market workflows.
Application	TanStack Router, TanStack Query, reusable components, server functions, loading states, and cached queries.
Data	Supabase authentication, PostgreSQL persistence, user-scoped records, roadmap progress, MCQ attempts, and notifications.
Intelligence	Multi-provider AI generation for diagnosis, roadmaps, market analysis, recruiter feedback, and interview practice.

7. What Has Been Built

CareerPilot is a working, deployed prototype rather than a concept-only proposal. The implemented product includes authentication, onboarding, target role selection, Pakistani and international company selection, custom company entry, career diagnosis, market analysis, personalized roadmaps, daily tasks, MCQ assessments, interview preparation, recruiter simulation, resume analysis, notifications, progress tracking, certificate unlocking, and mobile-responsive layouts.

8. Impact and Feasibility

CareerPilot can reduce wasted learning time, improve student confidence, and make career guidance more accessible. For universities and training organizations, it provides a structured employability journey. The architecture is modular, cloud-ready, and capable of expanding with more roles, employers, market sources, learning paths, and assessment content.

9. Demonstration Flow

A judge can understand the complete value of CareerPilot through this short product journey:

- Create or open a student profile.
- Choose a target role and company, including a custom company.
- Run a career diagnosis and review the rejection risks.
- Open Market Reality to see what the selected role requires.
- Generate the personalized roadmap.
- Open a roadmap day and complete the practical task.
- Take the role-specific MCQ and review correct answers and explanations.
- Practice an interview question and ask the recruiter simulator for feedback.
- Review progress, notifications, and certificate completion status.

10. Why CareerPilot Matters

CareerPilot responds to a practical national challenge: helping educated young people become employable through better information, focused learning, and credible evidence. It supports students who do not know where to begin and gives them a way to understand their profile from an employer's perspective.

CareerPilot AI helps every student understand the gap between their current profile and the job they want, then gives them a practical path to close that gap with real evidence.

